

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456263.6328 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.098 +/- 0.032
Transit depth (Rp/Rs)^2	0.95 +/- 0.62 [%]
Orbital Inclination (inc)	87.82 +/- 1.38
Airmass coefficient 1 (a1)	0.9877 +/- 0.0095
Airmass coefficient 2 (a2)	0.012 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	#3 - [375, 132]
Optimal Aperture	3.75
Optimal Annulus	10.97
Transit Duration (day)	0.1304 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.098 ± 0.032 vs 0.1036 (deviation 0.08σ)
Duration fit vs published	0.1304 d vs 0.1567 d (deviation 2.4σ)
Mid-time O-C	11.4 min
Depth SNR	1.4
Residual scatter	0.96 %

Light curve

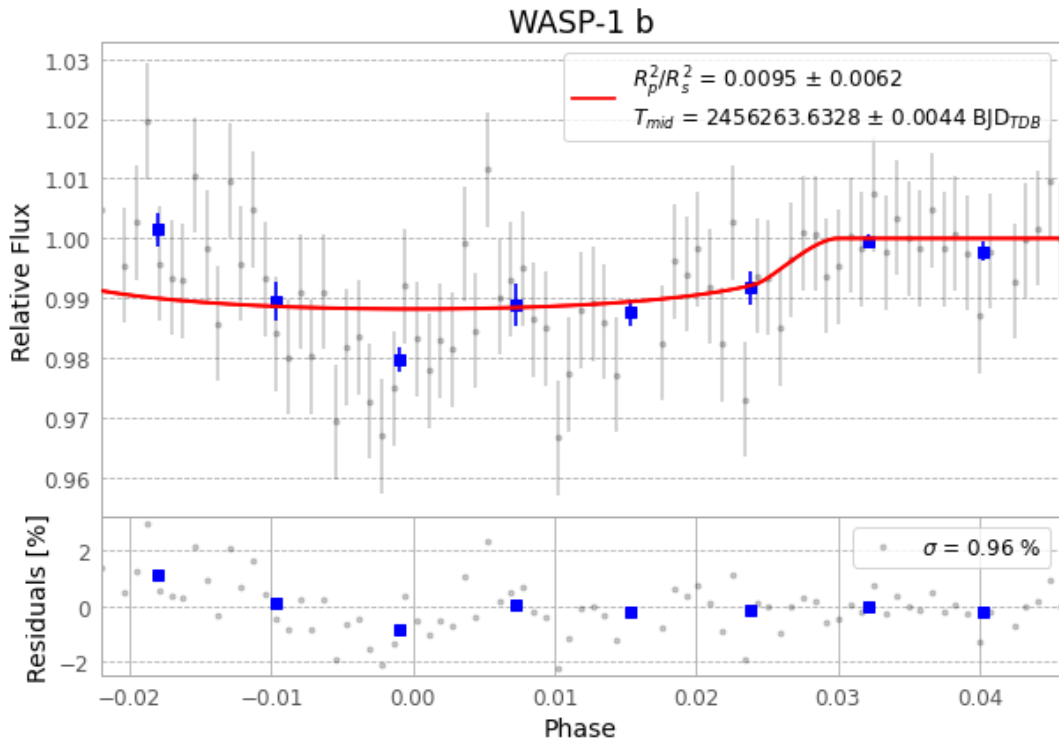


Figure 1 — FinalLightCurve_WASP-1 b_2012-12-01.png (SHA-256 86ae0907e326189c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8b04bf0ff68820...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

82 FITS frames (54 MB), WASP-1121202014512.FITS → WASP-1121202055112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121202.log (be4a8dbc0ba5...), FinalLightCurve_WASP-1 b_2012-12-01.png (86ae0907e326...), FinalLightCurve_WASP-1 b_2012-12-01.csv (638d4659bf80...)

Compute resources

Wall time 212.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:32Z.